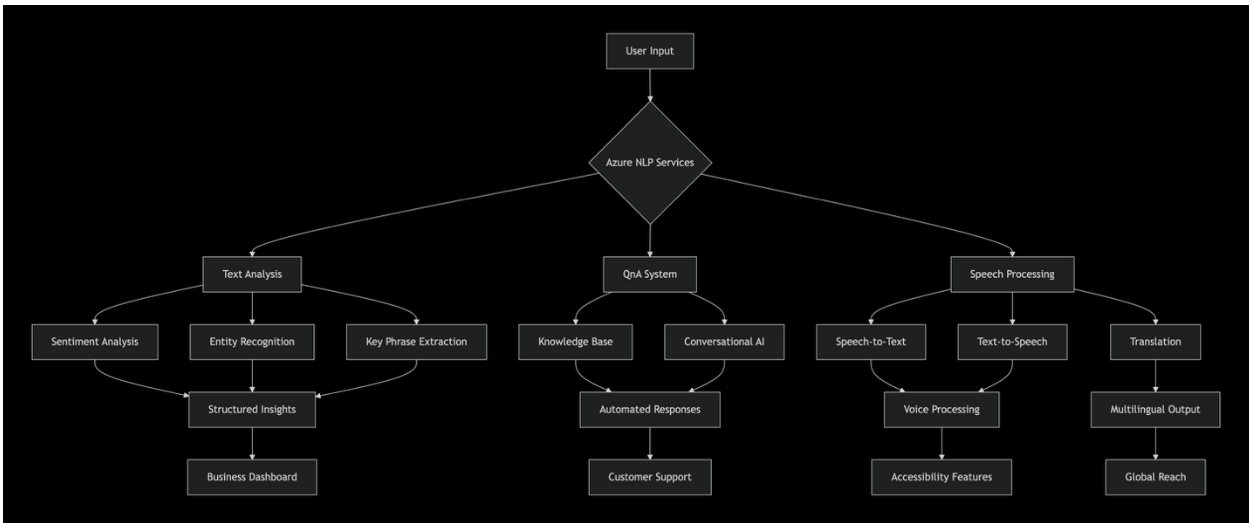
ANASIEZE IKENNA – CLOUD & AI ENGINEER

Project: Azure-AI Powered Natural Language Processing (NLP) Solutions

Overview

I implemented an Azure Cognitive Services NLP solution that automates text and voice processing, delivers real-time insights, and enables conversational AI at scale. This project demonstrates expertise in cloud-native AI, secure integration, and business value delivery using Microsoft Azure.

Architecture



Problem

Working with raw image data locally often results in:

- Manual tagging and annotation that doesn't scale
- Limited access to advanced computer vision without complex setups
- Risk of inconsistent results using offline tools
- Slow workflows that delay insight extraction

Goal: Create a lightweight, reproducible tool that users can use to:

- Analyze images using pre-trained and custom models

Tech Stack

Component	Tools Used
Programming	Python 3.10

SDKs	Azure AI Language SDK · Azure AI Speech SDK
NLP Services	Sentiment Analysis · Custom Classification · CLU (Conversational Language Understanding) · QnA Maker · Translator
Security	dotenv for credentials
Component	Tools Used
Outputs	JSON logs · Annotated text · Conversational bot prototype

Problem Statement

Organizations face challenges with:

- High volumes of unstructured text and voice data.
- Repetitive FAQs consuming support bandwidth.
- Manual analysis that is time-consuming and error-prone.
- Inconsistent multilingual accessibility for global users.

Objectives

- Build an AI-powered conversational system using Azure CLU + QnA.
- Automate text analytics for sentiment, classification, and entity recognition.
- Integrate speech-to-text and text-to-speech for accessibility.
- Enable multilingual workflows with Azure Translator.
- Ensure deployment is secure, modular, and scalable.

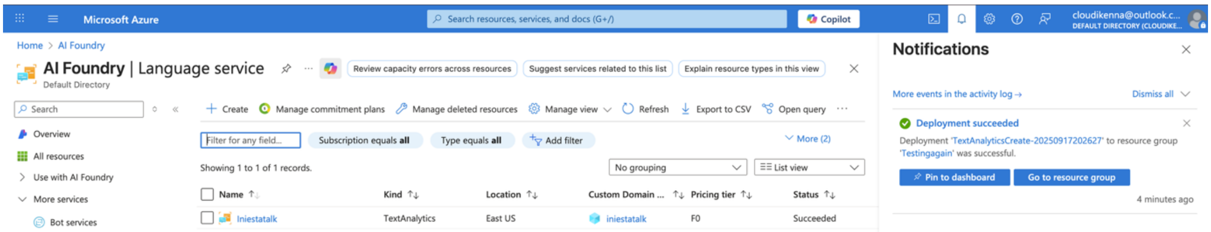
Implementation Steps

- Provision Azure Resources – Language, Speech, CLU, and QnA.
- Sentiment & Classification – Analyzed sample data for tone and categories.
- Entity Recognition – Extracted domain-specific terms (IDs, departments).
- Conversational AI – Configured CLU intents + QnA knowledge base.
- Speech Integration – Enabled speech-to-text and text-to-speech.
- Testing & Validation – Ran sample queries and voice inputs to confirm outputs.

- **Set Up**

⇒ **Text Analysis Application**

- *Provision an Azure AI Language resource*



- *Install requirements & Dependencies*

```
PS /home/ikenna/mslearn-ai-language/Labfiles/01-analyze-text/Python/text-analysis>  python -m venv labenv
PS /home/ikenna/mslearn-ai-language/Labfiles/01-analyze-text/Python/text-analysis>  ./labenv/bin/Activate.ps1
(labenv) PS /home/ikenna/mslearn-ai-language/Labfiles/01-analyze-text/Python/text-analysis>  pip install -r requirements.txt azure-ai-textanalytics==5.3.0
Collecting azure-ai-textanalytics==5.3.0
  Downloading azure_ai_textanalytics-5.3.0-py3-none-any.whl.metadata (82 kB)
Collecting python-dotenv (from -r requirements.txt (line 1))
  Downloading python_dotenv-1.1.1-py3-none-any.whl.metadata (24 kB)
Collecting azure-core<2.0.0,>=1.24.0 (from azure-ai-textanalytics==5.3.0)
  Downloading azure_core-1.35.1-py3-none-any.whl.metadata (46 kB)
Collecting azure-common~=1.1 (from azure-ai-textanalytics==5.3.0)
  Downloading azure_common-1.1.28-py2.py3-none-any.whl.metadata (5.0 kB)
Collecting isodate<1.0.0,>=0.6.1 (from azure-ai-textanalytics==5.3.0)
  Downloading isodate-0.7.2-py3-none-any.whl.metadata (11 kB)
Collecting typing-extensions>=4.0.1 (from azure-ai-textanalytics==5.3.0)
  Downloading typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
```

Run Application Code

- Configure variables **.env**
- Run **text-analysis.py**

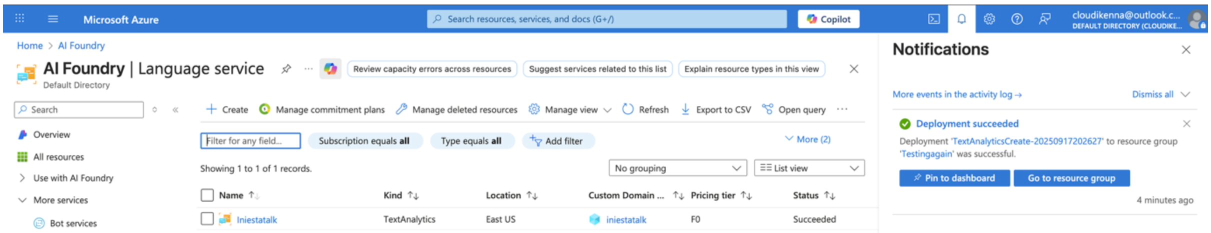
Outputs

```
review2.txt

Tired hotel with poor service
The Royal Hotel, London, United Kingdom
5/6/2018
This is a old hotel (has been around since 1950's) and the room furnishings are average - becoming a bit old now and require changing. The internet didn't work and had to come to one of their office rooms to check in for my
 flight home. The website says it's close to the British Museum, but it's too far to walk.

Language: English

Sentiment: negative

Key Phrases:
        The Royal Hotel
        Tired hotel
        old hotel
        poor service
        United Kingdom
        room furnishings
        office rooms
        flight home
        British Museum
        London
        changing
        internet
        website
        1950

Entities
        hotel (Location)
        The Royal Hotel (Organization)
        London (Location)
        United Kingdom (Location)
        5/6/2018 (DateTime)
        hotel (Location)
        since 1950's (DateTime)
        furnishings (Product)
        now (DateTime)
        internet (Skill)
        one (Quantity)
        office rooms (Location)
        flight (Event)
        home (Location)
        British Museum (Location)

Links
        The Royal Hotel (https://en.wikipedia.org/wiki/The_Royal_Hotel)
        London (https://en.wikipedia.org/wiki/London)
        British Museum (https://en.wikipedia.org/wiki/British_Museum)
```

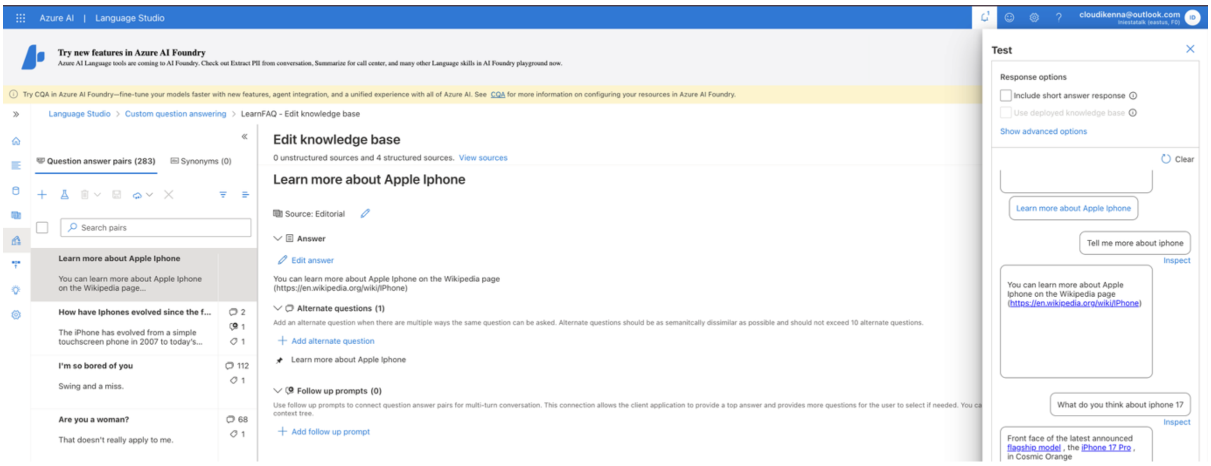
```
Good Hotel and staff
The Royal Hotel, London, UK
3/2/2018
Clean rooms, good service, great location near Buckingham Palace and Westminster Abbey, and so on. We thoroughly enjoyed our stay. The courtyard is very peaceful and we went to a restaurant which is part of the same group a
nd is Indian ( West coast so plenty of fish) with a Michelin Star. We had the taster menu which was fabulous. The rooms were very well appointed with a kitchen, lounge, bedroom and enormous bathroom. Thoroughly recommended.

Language: English

Sentiment: positive

Key Phrases:
        The Royal Hotel
        Good Hotel
        good service
        great location
        Buckingham Palace
        Westminster Abbey
        same group
        West coast
        Michelin Star
        taster menu
        enormous bathroom
        Clean rooms
        staff
```

⇒ Question-answering solution

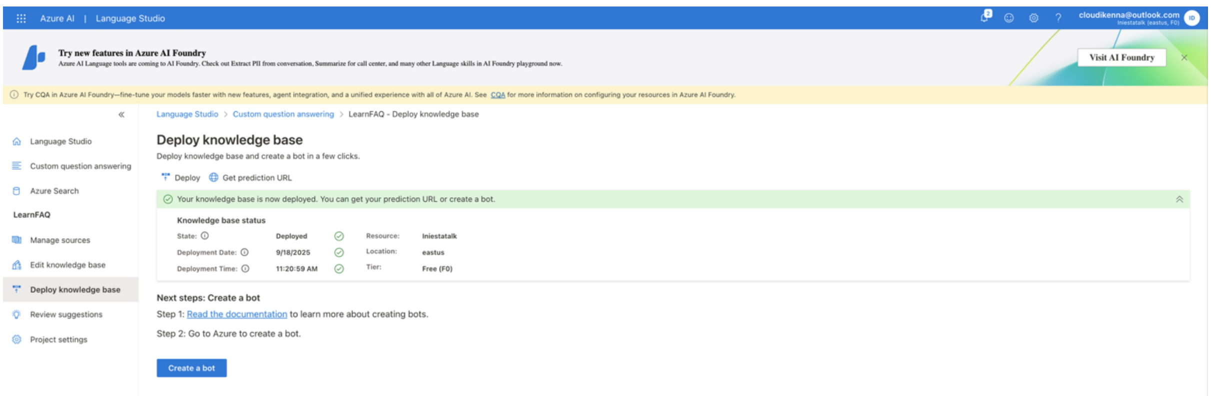
- *Provision an Azure AI Language resource*



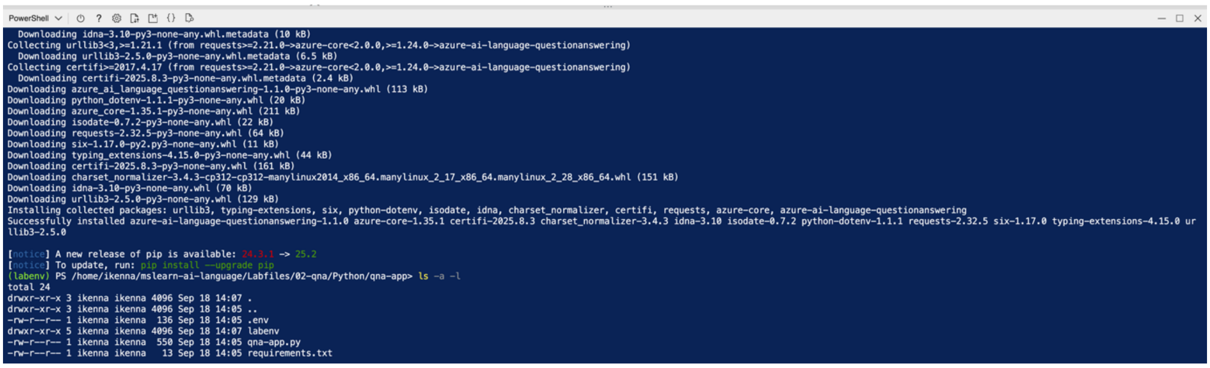
- *Configure, Test & Train Knowledge Base.*



- *Deploy knowledge base*



- *Install requirements & Dependencies*

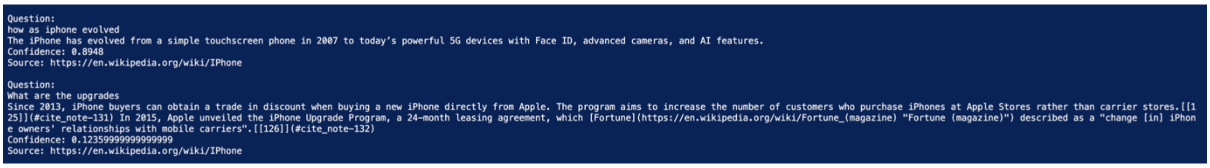


```
PowerShell
  Downloading idna-3.10-py3-none-any.whl.metadata (10 kB)
Collecting urllib3<3,>=1.21.1 (from requests>=2.21.0->azure-core<2.0.0,>=1.24.0->azure-ai-language-questionanswering)
  Downloading urllib3-2.5.0-py3-none-any.whl.metadata (6.5 kB)
Collecting certifi>=2017.4.17 (from requests>=2.21.0->azure-core<2.0.0,>=1.24.0->azure-ai-language-questionanswering)
  Downloading certifi-2025.8.3-py3-none-any.whl.metadata (2.4 kB)
Downloading azure_ai_language_questionanswering-1.1.0-py3-none-any.whl (113 kB)
Downloading python_dotenv-1.1.1-py3-none-any.whl (20 kB)
Downloading azure_core-1.35.1-py3-none-any.whl (211 kB)
Downloading isodate-0.7.2-py3-none-any.whl (22 kB)
Downloading requests-2.32.5-py3-none-any.whl (64 kB)
Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
Downloading typing_extensions-4.15.0-py3-none-any.whl (44 kB)
Downloading certifi-2025.8.3-py3-none-any.whl (161 kB)
Downloading charset_normalizer-3.4.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (151 kB)
Downloading idna-3.10-py3-none-any.whl (70 kB)
Downloading urllib3-2.5.0-py3-none-any.whl (129 kB)
Installing collected packages: urllib3, typing-extensions, six, python-dotenv, isodate, idna, charset_normalizer, certifi, requests, azure-core, azure-ai-language-questionanswering
Successfully installed azure-ai-language-questionanswering-1.1.0 azure-core-1.35.1 certifi-2025.8.3 charset_normalizer-3.4.3 idna-3.10 isodate-0.7.2 python-dotenv-1.1.1 requests-2.32.5 six-1.17.0 typing-extensions-4.15.0 ur
llib3-2.5.0

[notice] A new release of pip is available: 24.3.1 -> 25.2
[notice] To update, run: pip install --upgrade pip
(labenv) PS /home/ikenna/mslearn-ai-language/Labfiles/02-qna/Python/qna-app> ls -a -l
total 24
drwxr-xr-x 3 ikenna ikenna 4096 Sep 18 14:07 .
drwxr-xr-x 3 ikenna ikenna 4096 Sep 18 14:05 ..
-rw-r--r-- 1 ikenna ikenna  136 Sep 18 14:05 .env
drwxr-xr-x 5 ikenna ikenna 4096 Sep 18 14:07 labenv
-rw-r--r-- 1 ikenna ikenna  550 Sep 18 14:05 qna-app.py
-rw-r--r-- 1 ikenna ikenna   13 Sep 18 14:05 requirements.txt
```

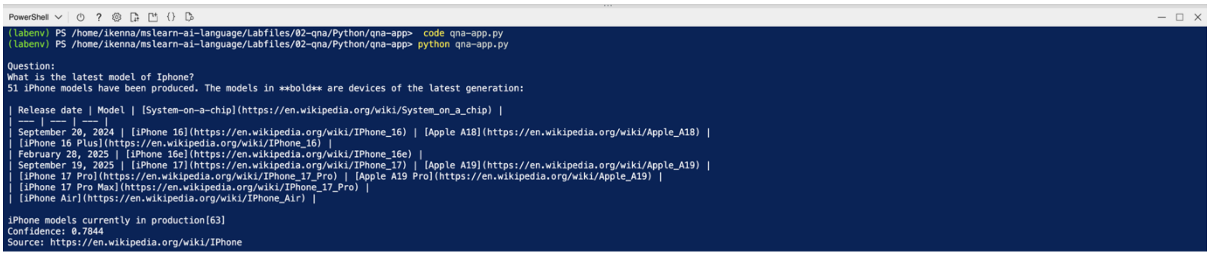
Configure Application

- Configure variables **.env**
- Run **qna-app.py**

Outputs



```
Question:
how as iphone evolved
The iPhone has evolved from a simple touchscreen phone in 2007 to today's powerful 5G devices with Face ID, advanced cameras, and AI features.
Confidence: 0.8948
Source: https://en.wikipedia.org/wiki/IPhone

Question:
What are the upgrades
Since 2013, iPhone buyers can obtain a trade in discount when buying a new iPhone directly from Apple. The program aims to increase the number of customers who purchase iPhones at Apple Stores rather than carrier stores.[[1
25]](#cite_note-131) In 2015, Apple unveiled the iPhone Upgrade Program, a 24-month leasing agreement, which [Fortune](https://en.wikipedia.org/wiki/Fortune_(magazine) "Fortune (magazine)") described as a "change [in] iPhon
e owners' relationships with mobile carriers".[[126]](#cite_note-132)
Confidence: 0.12359999999999999
Source: https://en.wikipedia.org/wiki/IPhone
```



```
PowerShell
(labenv) PS /home/ikenna/mslearn-ai-language/Labfiles/02-qna/Python/qna-app>  code qna-app.py
(labenv) PS /home/ikenna/mslearn-ai-language/Labfiles/02-qna/Python/qna-app> python qna-app.py

Question:
What is the latest model of Iphone?
51 iPhone models have been produced. The models in **bold** are devices of the latest generation:

| Release date | Model | [System-on-a-chip](https://en.wikipedia.org/wiki/System_on_a_chip) |
| --- | --- | --- |
| September 20, 2024 | [iPhone 16](https://en.wikipedia.org/wiki/IPhone_16) | [Apple A18](https://en.wikipedia.org/wiki/Apple_A18) |
| [iPhone 16 Plus](https://en.wikipedia.org/wiki/IPhone_16) |
| February 28, 2025 | [iPhone 16e](https://en.wikipedia.org/wiki/IPhone_16e) |
| September 19, 2025 | [iPhone 17](https://en.wikipedia.org/wiki/IPhone_17) | [Apple A19](https://en.wikipedia.org/wiki/Apple_A19) |
| [iPhone 17 Pro](https://en.wikipedia.org/wiki/IPhone_17_Pro) | [Apple A19 Pro](https://en.wikipedia.org/wiki/Apple_A19) |
| [iPhone 17 Pro Max](https://en.wikipedia.org/wiki/IPhone_17_Pro) |
| [iPhone Air](https://en.wikipedia.org/wiki/IPhone_Air) |

iPhone models currently in production[63]
Confidence: 0.7844
Source: https://en.wikipedia.org/wiki/IPhone
```

Key Outcomes

- 5X Faster query handling with automated FAQs.
- Structured insights from raw text (sentiment, categories, entities).
- Multilingual + voice support for accessibility.
- Secure, scalable deployment leveraging Azure Cognitive Services.

Business Value

- Faster and more consistent customer support.
- Reduced operational workload for teams.
- Global reach with translation and voice support.
- Actionable insights from text that drive decision-making.

This project highlights my skills in ***Azure AI, NLP, and cloud deployment*** *— showing how I design and implement real-world AI solutions for business workflows.*